

Introduction to Kernel Density Estimation

Table of contents

Density estimation	2
Basic Idea of kernel density estimation	5
Multivariate nonparametric density estimation	14
Homework	15

Density estimation

In probability and statistics, density estimation is the construction of an estimate, based on observed data, of an unobservable underlying probability density function.

Distribution

1. Age: 23, 28, 30,
2. Event: 1, 0, 0, 1,
3. The number of defects in a time period: 0, 1, 4, 4, 8.....

Old Faithful Geyser

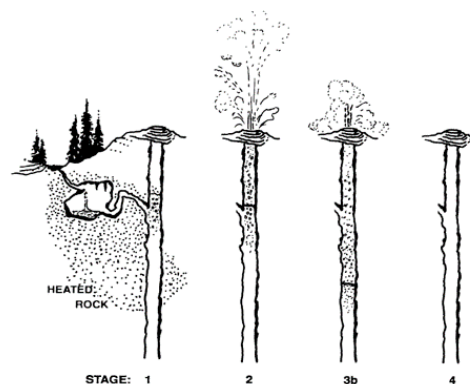


Fig. 5. Sketch of the conjectured geyser mechanism (from Rinehart (1969))

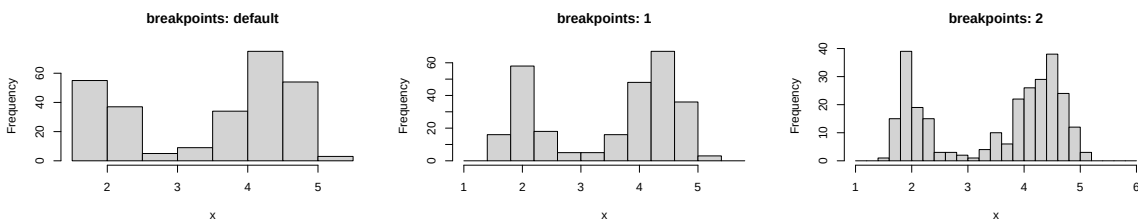
An analysis of data on the duration times and waiting times for eruptions. Times between Old Faithful eruptions (Y) and duration of eruptions (X).

Old Faithful Geyser

```
data(faithful)
attach(faithful)
str(faithful)
```

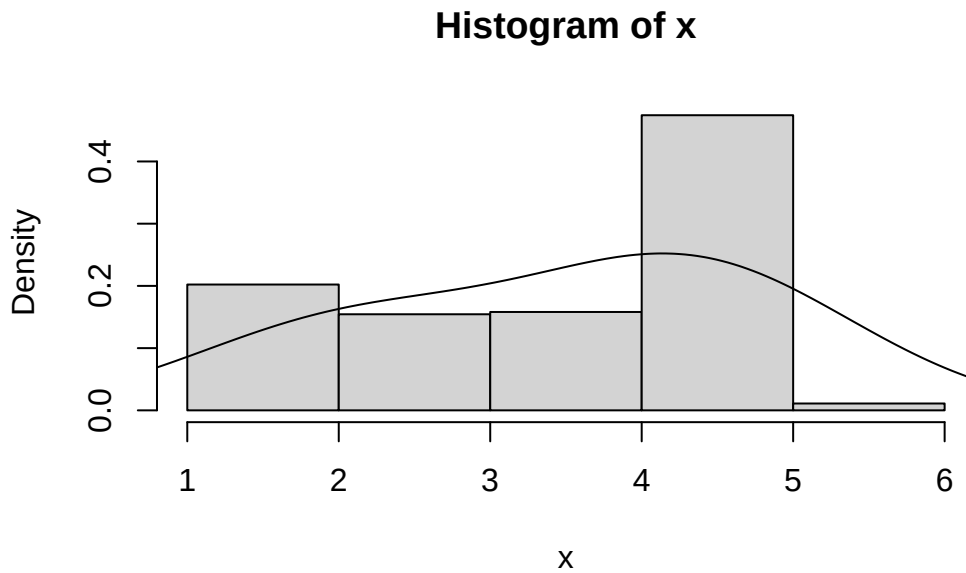
```
'data.frame':  272 obs. of  2 variables:
 $ eruptions: num  3.6 1.8 3.33 2.28 4.53 ...
 $ waiting  : num  79 54 74 62 85 55 88 85 51 85 ...
```

```
x <- eruptions
hist(x, main="breakpoints: default")
hist(x, breaks=seq(1, 6, 0.4), main="breakpoints: 1")
hist(x, breaks=seq(1, 6, 0.2), main="breakpoints: 2")
```



Bandwidth

```
x <- eruptions
hist(x, breaks=4, freq = F)
lines(density(x, bw = 1))
```



- Using the above diagram as an example, the overall range of the data can be represented by a bar chart.
- Currently, it is divided into 5 bins, and the number of data points within each bin can be calculated and denoted as n_i .
- The probability of that bin can be expressed as:

$$p_i = \frac{n_i}{\sum_{i=1} n_i} = \frac{n_i}{N}$$

Histogram-based method

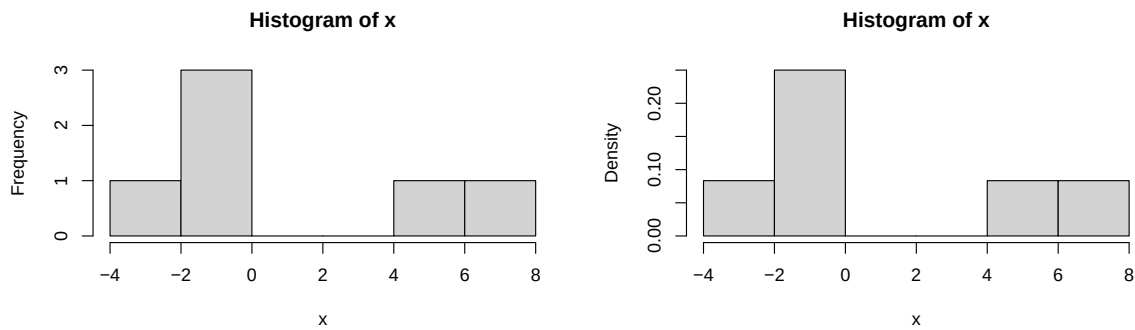
The simplest form of non-parametric D.E. is the familiar histogram

$$f(x) = \frac{1}{N} \frac{\text{number of } x \text{ in the bin}}{\text{width of bin}}$$

```
x <- c(-2.1, -1.3, -0.4, -1.9, 5.1, 6.2)
```

```
hist(x, freq = T)
hist(x, freq = F)
```

The density for the interval $[-2, 0]$.



3/(6*2)

[1] 0.25

Histogram-based method

The histogram is a very simple form of D.E., but it has several drawbacks

1. The final shape of the density estimate depends on the starting position of the bins.
2. The discontinuities of the estimate are not due to the underlying density, they are only an artifact of the chosen bin locations.
3. A much more serious problem is the curse of dimensionality, since the number of bins grows exponentially with the number of dimensions.

Basic Idea of kernel density estimation

- In nonparametric statistics, no parametric assumptions are made on the underlying probability model that generated the data. Nonparametric kernel smoothing methods are often, however, much more computationally demanding.
- The classical approach to estimating the probability density that generated an observed sample is to assume a parametric model. But if the assumed parametric model is not correct, inferences derived from it can lead to misleading interpretations.
- In nonparametric density estimation, the resulting density estimator is more flexible and the data are “allowed to speak for themselves”. This results in a powerful tool for exploratory data analysis: visualization of data sets, classification

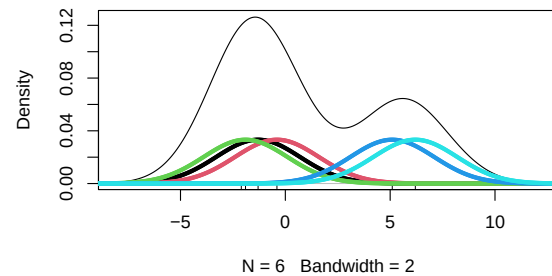
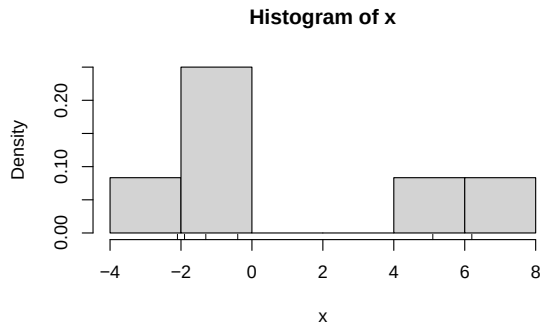
Drawbacks

- The usual estimators depend on a smoothing parameter, typically hard to select: there is no unique obvious optimal choice for it.
- Need for large samples: the theoretical motivation for the estimators is usually asymptotic. The estimators usually have a poor performance if the sample size is small.
- The curse of dimensionality: difficulties in higher dimensions - as the data dimension grows the estimators require larger and larger sample sizes.

A simple example

Suppose that we have a dataset:

Sample ID	1	2	3	3	5	6
value	-2.1	-1.3	-0.4	-1.9	5.1	6.2



Kernel function

A kernel is a [non-negative real-valued integrable](#) function K . For most applications, it is desirable to define the function to satisfy two additional requirements:

- [Normalization](#):

$$\int_{-\infty}^{+\infty} K(u) du = 1;$$

- [Symmetry](#):

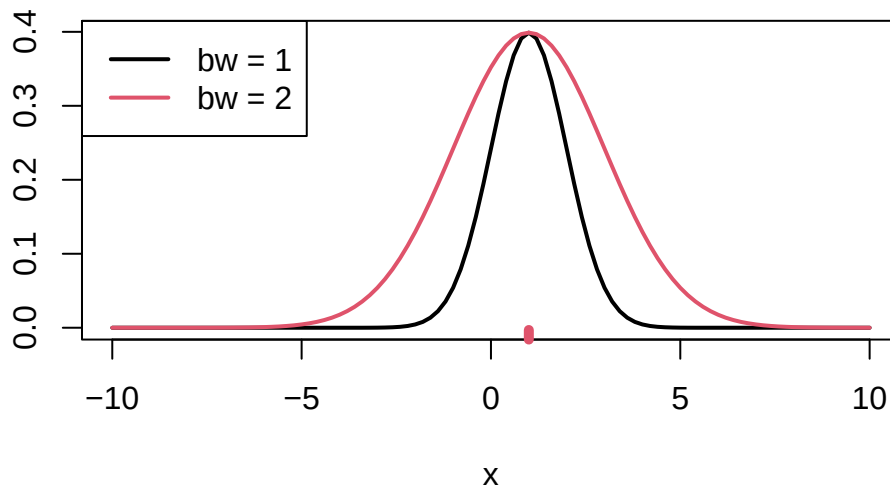
$$K(-u) = K(u) \text{ for all values of } u.$$

The first requirement ensures that the method of kernel density estimation results in a [probability density function](#). The second requirement ensures that the average of the corresponding distribution is equal to that of the sample used.

If K is a kernel, then so is the function K^* defined by $K^*(u) = \lambda K(\lambda u)$, where $\lambda > 0$. This can be used to select a scale that is appropriate for the data.

Figure 1: Source: Wiki

Kernel function



- $K(x - x_i)$: Take a single data point x_i belonging to our dataset X .
- $K(\frac{x - x_i}{h})$: To make the curve wider or narrower we can just throw a constant h .
- $\frac{1}{h}K(\frac{x - x_i}{h})$: The area under the kernel function is multiplied by h as a result. Therefore, we have to restore it back to the unit area by dividing by h .

$$\int_{-\infty}^{\infty} K\left(\frac{x-x_i}{h}\right)dx = h$$

Theory

A simple density estimator: Let $a_0 < a_1 < \dots < a_i < a_{i+1}$ with $h_{i+1}^n = a_{i+1}^n - a_i^n$. The empirical density function can be calculated as

$$\hat{f}_n^t = \frac{\{\#i : X_i \in (a_j, a_{j+1})\}}{nh_n}$$

Kernel estimators, definition and motivation

This is a moving window estimator

$$\hat{f}_n(x) = \frac{\sum_{i=1}^n I_{[x_i-0.5, x_i+0.5]}(x)}{n} = \frac{1}{n} \frac{1}{1} \sum_{i=1}^n U\left(\frac{x-x_i}{1}\right)$$

This can be generalized by replacing the uniform density with another K (call kernel)

$$\hat{f}_n(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h_n}\right)}{nh_n}$$

where $h_n > 0$ is a smoothing parameter and called bandwidth.

Note

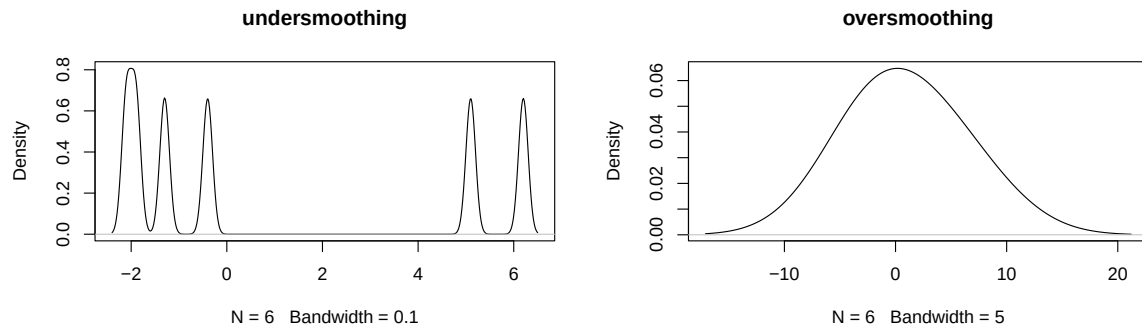
This is a smoothed version of the histogram.

The Bandwidth

The selected bandwidth controls how smooth the estimated density curve is. There are some convenient things to remember about the bandwidth:

- As we increase bandwidth, points further from our location are included. This results in a smoother distribution.
- Too large of a bandwidth can lead to the loss of information about the shape of the underlying data. This is called oversmoothing.
- Too small of a bandwidth may result in the density reflecting randomness rather than our true underlying density. This is called undersmoothing.

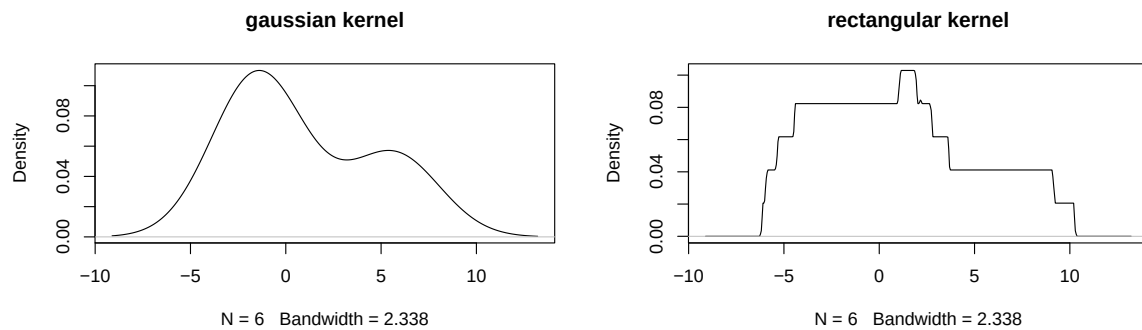
The Bandwidth



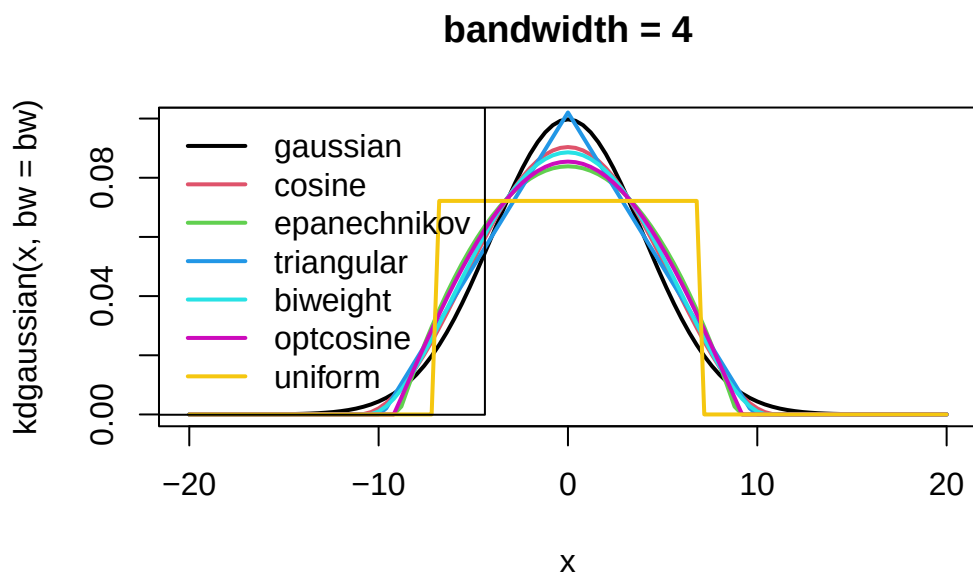
Possible kernel choice

The possible kernel choice are gaussian, epanechnikov, rectangular, triangular, biweight, cosine, optcosine.

```
plot(density(x, kernel = "gaussian"), main="gaussian kernel")
plot(density(x, kernel = "rectangular"), main="rectangular kernel")
```



Possible kernel choice



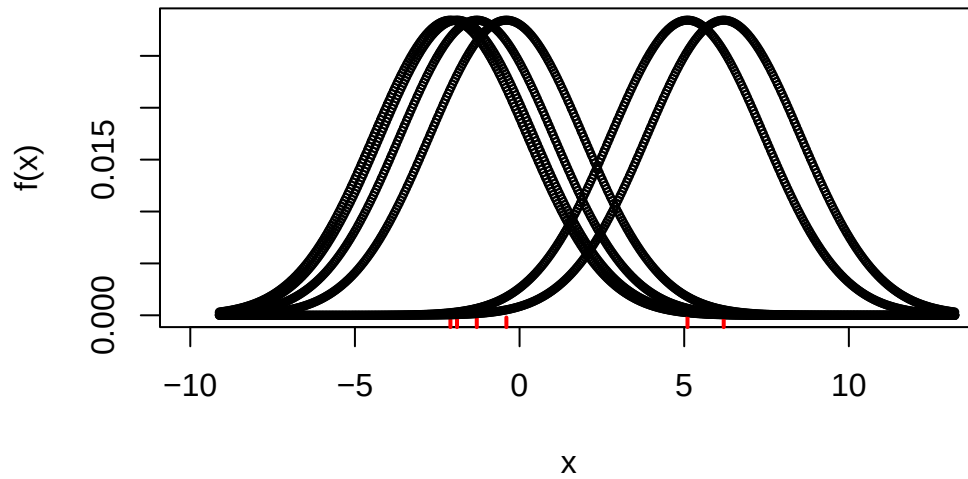
Construct K.D. function step by step

```
x <- c(-2.1, -1.3, -0.4, -1.9, 5.1, 6.2)
myt <- density(x, kernel = "gaussian")$x
fivenum(myt)
```

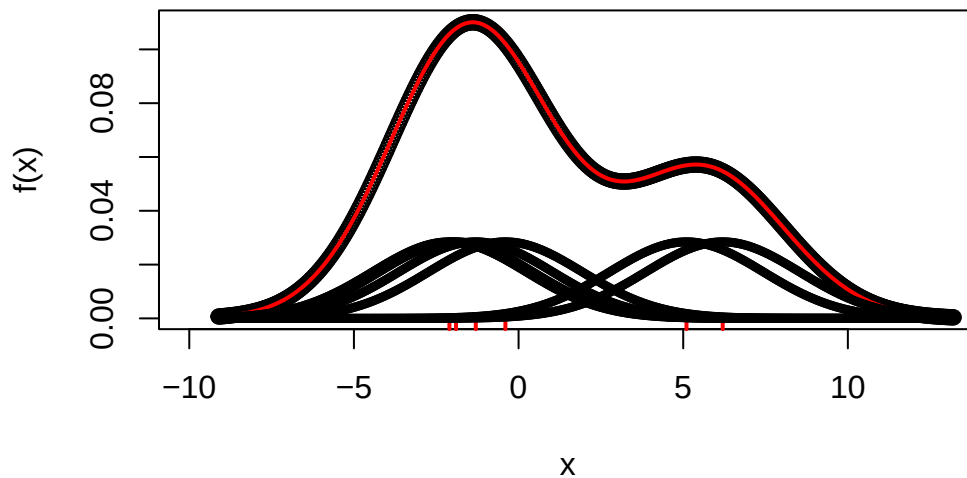
```
[1] -9.114013 -3.542930 2.050000 7.642930 13.214013
```

```
n <- length(x)
bw <- 2.338
ft <- matrix(NA, ncol=length(myt), nrow = n)
for (i in 1:n){
  ft[i, ] <- dnorm( (myt-x[i])/bw )/(n*bw)
}
plot(x = myt, y = ft[1,], xlim = c(-10,13), cex=0.5, xlab="x", ylab="f(x)")
rug(x, lwd=2, col="red")
points(x = myt, y = ft[2,], xlim = c(-10,13), cex=0.5)
points(x = myt, y = ft[3,], xlim = c(-10,13), cex=0.5)
```

```
points(x = myt, y = ft[4,], xlim = c(-10,13), cex=0.5)
points(x = myt, y = ft[5,], xlim = c(-10,13), cex=0.5)
points(x = myt, y = ft[6,], xlim = c(-10,13), cex=0.5)
```



```
yhat <- colSums(ft)
plot(x = myt, y = yhat, xlim = c(-10,13), cex=1, type = "p", col="black", xlab="x", ylab="f(x)")
rug(x, lwd=2, col="red")
lines(density(x, kernel = "gaussian"), col="red", lwd=2)
for (i in 1:6) points(x = myt, y = ft[i,], xlim = c(-10,13), cex=0.5)
```

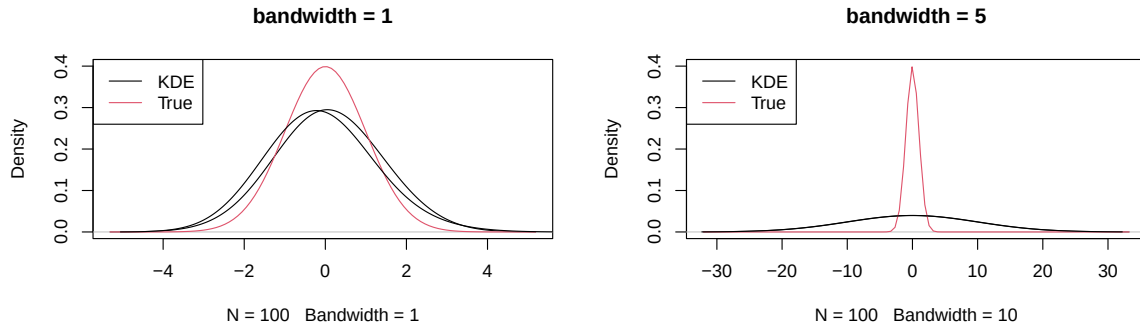


Criteria for the estimation error

```
set.seed(123)
x1 <- rnorm(100)
x2 <- rnorm(100)
plot(density(x1, bw = 1), main="bandwidth = 1", ylim=c(0,0.4))
curve(dnorm, add = T, col=2)
lines(density(x2, bw= 1))
legend("topleft",c("KDE","True"),lty=1, col=c(1,2))
plot(density(x2, bw =10), main="bandwidth = 5",ylim=c(0, 0.4))
curve(dnorm, add = T, col=2)
lines(density(x1, bw =10))
legend("topleft",c("KDE","True"),lty=1, col=c(1,2))
```

i Note

1. A large bandwidth will reduce the differences among the estimates of $p_{KDE}(x)$ for different data sets, but it will increase the bias of $p_{KDE}(x)$ with respect to the true density.
2. small bandwidth will reduce the bias of $p_{KDE}(x)$, at the expense of a larger variance



in the estimates $p_{KDE}(x)$.

Bandwidth selection methods

Reference to a standard distribution

- Assume a standard density function and find the value of the bandwidth that minimizes the integral of the square error (MISE)

$$h_{opt} = \operatorname{argmin}_h \{MISE(P_{KDE}(x))\}$$

where $MISE(P_{KDE}(x))$ is

$$E \left[\int (P_{KDE}(x) - f(x))^2 dx \right]$$

and $f(x)$ is the standard density function.

Bandwidth selection methods

Likelihood cross-validation

- It is to maximize the “pseudo-likelihood” computed using leave-one-out cross-validation

$$h_{MLCV} = \operatorname{argmax}_h \left\{ \frac{1}{N} \sum_{n=1}^N \log P_{-n}(x_n) \right\}$$

where $P_{-n}(x_n) = \frac{1}{(N-1)h} \sum_{m=1, m \neq n}^N K\left(\frac{x^{(n)} - x^{(m)}}{h}\right)$.

Bandwidth Selectors for Kernel Density Estimation in R

```
x <- c(-2.1, -1.3, -0.4, -1.9, 5.1, 6.2)
# help(bw.nrd)
# the default setting in density function
bw.nrd(x)
```

```
[1] 2.338004
```

```
density(x)$bw
```

```
[1] 2.338004
```

```
# unbiased cross-validation
bw.ucv(x)
```

```
[1] 1.344187
```

```
# biased cross-validation
bw.bcv(x)
```

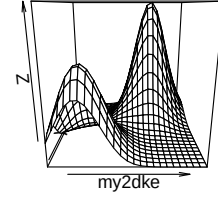
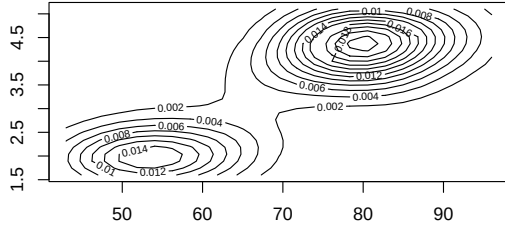
```
Warning in bw.bcv(x): minimum occurred at one end of the range
```

```
[1] 2.960023
```

Multivariate nonparametric density estimation

It is straightforward to extend the idea of kernel density estimation to obtain multidimensional densities. There exist a number of packages that can perform multivariate kernel density estimation in R. Times between Old Faithful eruptions (Y) and duration of eruptions (X).

```
library(MASS)
my2dke <- kde2d(faithful$waiting, faithful$eruptions)
contour(my2dke)
persp(my2dke)
```



Homework

Question 1: Use the data set `airquality` in R to answer the following questions:

1. Find the correlation matrix for Ozone, Solar R, Wind, and Temperature by excluding missing values.
2. Extract “Ozone” data vector. Plot the histogram with a **step-by-step** kernel density estimate.
3. Based on question (2), use gamma distribution to fit the dataset “Ozone” and plot it. What’s the estimations of the shape and scale parameters? Does it close to the kernel density plot?
4. Please show time series plots based on daily readings of Ozone, Solar R, Wind, and Temperature for May 1 to September 30. What is your finding?

Question 2: Let the kernel function $K(x)$ is a probability density symmetric around zero. Show that a kernel density estimator

$$\hat{f}_n(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h_n}\right)}{nh_n}$$

is a probability density function.